

STATISTICS IN SOUTH AFRICA*

J.E. Kerrich

University of the Witwatersrand

Of all people, we, the 180 members of the South African Statistical Association, should have most knowledge and clearest ideas on what I shall term the "state of statistics in South Africa".

When trying to clarify what I mean by "state of statistics", I find myself building up from the following basic facts.

- (i) Anyone who counts or measures is collecting statistics. Anyone who then tries to extract information from these counts or measures, is trying to analyse statistics and interpret the results.
- (ii) Very often the numerical data collected form part only of the totality of numerical data that we would like to examine. What does the part tell us about the whole? The age old problem of induction has once again cropped up and with it the vague concept of probability has reared its ugly head.

These basic facts have led to the creation of a body of specialists, call them trained statisticians, who in various ways and at various levels of sophistication are particularly skilled in methods of collecting, analysing, and interpreting numerical data, with special emphasis on data obtained by sampling.

In what follows I shall seldom mention those statisticians who, though highly skilled in some ways, lack formal training in the mathematical theory of statistics, but will talk mainly about those who have had some such mathematical training and will refer to them as mathematical statisticians.

In South Africa

- (a) How are mathematical statisticians trained and what are they taught?
- (b) Where are they employed?
- (c) What is their status with their employers?

I shall regard my (partial) answers to these questions as constituting a picture of the present state of statistics in South Africa.

In addition, it will be of interest to glance at the past and make some comparisons with what is happening in other countries.

Forty years ago, the bases of modern statistical theory had been truly laid (by K. Pearson, R.A. Fisher and others), but hardly anyone in this country appreciated their work, even at our Universities.

By the end of World War II, statistical methods were being used much more widely.

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At one University, mathematical statistics was accepted as part of the B.Sc. and B.A. curricula in 1954 and a chair of statistics was established in 1957. This year, some 80 undergraduates and 8 honours students are studying mathematical statistics there, together with 2 M.Sc. students and one Ph.D. student. This is somewhat similar to what occurred at the other Universities. Today chairs of statistics exist at Bloemfontein, Cape Town, Durban, Johannesburg (at two Universities), Potchefstroom, Pretoria (at two Universities) and Stellenbosch and courses in mathematical statistics are offered at Port Elizabeth and Rhodes. There is a chair of biometry at Pietermaritzburg. Some of these chairs were founded very recently. It augers well for the future of statistics in this country that they exist.

It seems that most of the mathematical statisticians are trained in the Faculty of Science at universities. I believe that in the main this is true all over the world.

What are they taught?

A Survey by our Executive in 1965 described undergraduate training in the various Faculties of Science as an attempt to provide in various ways a sound mathematical basis to most of the branches of modern statistical analysis together with some insight as to how they can be applied in practice.

What changes must we expect in these courses?

Between roughly 1930 and 1950, many advances were made in

- (i) Experimental Design
- (ii) Multivariate Analysis.

These have by now been well digested in various text books and many quite elementary texts have appeared containing portions of (i) and (ii) and differing considerably from the elementary texts of 20 years ago.

More recently, much research has been done in

- (iii) Nonparametric analysis.
- (iv) Stochastic processes, including queueing theory.
- (v) Games and decision theory.

How much of what we teach at present must we condense, simplify or omit to make room for what portions of (iii), (iv) or (v) in our undergraduate courses? Should we insist on an extra year of undergraduate study?

Our various honours courses deal mainly with topics (i) to (v) supplemented by some relevant topics from pure mathematics and mathematical probability, with emphasis at the discretion of the lecturers involved.

This is a wide and expanding field and there is a growing tendency to allow the overflow of new topics to seep into M.Sc. courses and award this rather high distinction on the results of more and more examinations and less and less research.

An interesting local academic development is that a University has just established an M.Sc. degree in Operations Research which involves three years of part-time study. The intention is to attract mathematicians, mathematical statisticians, engineers and mathematical economists into a field which is proving to be of increasing importance to administrators of large business concerns, etc. It is under the direction of the Department of Statistics of the University concerned. Analogous facilities exist overseas.

Courses in statistics are also offered in the Faculties of Commerce at many of our Universities, including the Universities of Bloemfontein, Cape Town, Potchefstroom, Pretoria, South Africa and Stellenbosch. They differ somewhat from B.Sc. courses in mathematical background and statistical emphasis and have rather different fields of application in mind. At some Universities they lead on to honours courses and higher degrees in statistics.

Statistics and statisticians are being profoundly influenced by the recent rapid development of electronic computers. I am sure that we still do not fully appreciate all the uses to which this tool can be put, but that as we understand these potentialities better, this knowledge will have an enormous effect on what we teach students (not only students of statistics) and how we teach it.

Quite recently at one University a Professor of Computing Science was appointed. He is also Director of a well equipped Computing Centre. Already this has had a strong influence on the students of Mathematical Statistics there.

To balance these glimpses of present and future developments, Table I gives a brief record of achievement in the immediate past. The table shows the number of postgraduate degrees awarded in statistics during the past five years, as reported by five S.A. Universities.

TABLE I

B.Sc. Honours	64	B.Comm. Honours	21
M.Sc.	15	M.Comm.	3
Ph.D.	6	D.Comm.	2
D.Sc.	<u>2</u>		<u>—</u>
	<u>87</u>		<u>26</u>

Three other universities reported nil under these headings because their courses were too recently established.

Turn now to the second question (b): Where are Mathematical Statisticians employed?

Our Universities were asked to try to trace all students who during the years 1964-65-66 had completed their third year undergraduate statistical courses and report on what they were doing now. The results handed in referred to full-time and part-time students in the Faculties of Science and Commerce. See Table II.

TABLE II

	<u>B.Sc.</u>		<u>B.Comm.</u>		<u>Total</u>
	Full- time	Part- time	Full- time	Part- time	
Lecturers (not all in Statistics)	9	5x)	1		15
Further full-time study (not always in Statistics)	43		5	5	53
Employed in:					
Research Institutes	8	8	3	2	21
Research Organisations serving coordinated industries.....	26	8	8	8	50
Research sections in individual firms	8	1		1	10
Organisations where Statistics is used occasionally	17				17
Computer work	15				15
Insurance offices etc. and studying for Actuarial qualifications ..	31		1		32
Teaching	19		5		24
Work not requiring Statistics	<u>6</u>	<u>3</u>	<u>6</u>	<u>—</u>	<u>15</u>
	<u>182</u>	<u>25</u>	<u>29</u>	<u>16</u>	<u>252</u>
Not traced	<u>72</u>	<u>38</u>	<u>25</u>	<u>3</u>	<u>138</u>
Total					<u>390</u>

Of those employed, at least 41 are continuing to study as part-time students, mainly in statistical topics.

x) Part-time students "now lecturing in Universities or Colleges".

The list does give an idea of the number of statisticians with some mathematical background and how they are employed.

Now consider my third question: (c) What is the status of mathematical statisticians with their employers?

A little background may be useful first.

During the thirties, mathematicians such as Kolmogoroff put probability on an axiomatic basis by writing down symbols p, q , etc., defined as mathematical probabilities, endowing them with certain mathematical properties and thereafter deducing long strings of mathematically correct theorems. This means that the pure probabilist is free to retire to his ivory tower and continue his mathematical deductions completely oblivious as to whether or not his results can be applied to practical problems. This leaves other people, call them applied statisticians, to

face the non-mathematical problem of judging which, if any, of these theorems can be regarded as a useful approximate model of a particular situation in physical nature. If so, then what sound inductive judgements can they make about this physical situation? Much literature and several conflicting schools of thought exist on how to make these practical judgements. The fact remains that these non-mathematical inductions are made and that pragmatically further experience often shows that they were reasonably correct.

One of the earliest works on our subject was *Ars Conjectandi*, by James Bernoulli, published in 1713. Since then our subject has developed enormously and in particular many valuable aids have been evolved to help us make sound judgements in the face of uncertainty. Nevertheless, in my opinion it should be realised with crystal clarity that to a very great extent it has remained and will remain *The Art of Guessing*.

An unfortunate result of the dichotomy of effort mentioned above is that the pure probabilist and the applied statistician find it more and more difficult to communicate with each other. This is frequently and properly commented on adversely. In our Association I think that both types of statistician are in fairly close contact with each other and I hope they will make every effort to continue to do so.

To return to the third question:-

Within a University, both types of statistician have their place, and have the status, I trust, to which their special capabilities entitle them.

Outside the University, it is the applied statistician who seeks employment. Here is a man trained in the collection, analysis, and interpretation of numerical data. To my mind, he MUST NOT merely be given data collected by other people and told to analyse them in some manner prescribed by his employers and thereafter to hand in the results of his analysis for them to interpret. His advice should be sought at every stage and the stages are inextricably intermingled: What is the problem; what data should be collected to throw light on it; how should the data be analysed; how do we interpret the results of the analysis? Of course the statistician has not got all the answers; if he has something to give, he has much to learn, but he should be part of a team of people, working together cordially and critically, where every member of team tells every other member as clearly as he can, exactly what he knows about the problem. In this way, they can together produce far better results than can be obtained by other means.

Is this properly understood by those who employ applied statisticians? In many cases I fear not. In any large organisation there has to be a hierarchy. Cordial and critical cooperation does not flourish easily in such an atmosphere.

I plead with all employers to use their statisticians intelligently and make full use of their training. Use a man to his full, and his proper status (whatever that may be) will follow.

Finally, how do problems concerned with organising the teaching and use of statistics in this country, compare with those of other countries? In my experience, their problems may differ in magnitude but are very similar in other respects. In 1949, Gertrude Cox lectured to the Royal Statistical Society on the

problems she met with in running her macroscopic Institute in North Carolina. During the discussion, I matched her, point by point, with the problems I met with (on a microscopic scale) as the only lecturer on mathematical statistics at my University.

In August, 1967, I attended two international statistical conferences in Sydney and discussed problems of this type with many of the other delegates, with the same result. It cheered me up to find that so many others have the same sort of problems as we have and don't know the exact answers either.

In October 1967, I presided over our own small annual conference and am pleased to say that in my opinion the quality of the papers read here matched the average quality of the papers read at the international conferences.

Here then is my picture (however incomplete) of Statistics in South Africa and the problems associated with it. If the picture is vague, that is partly due to the fact that the subject of this paper is in a state of rapid, healthy and rather untidy growth. Personally, I am delighted with this growth and am very reasonably satisfied with the standards that we have attained and will, I believe, maintain.

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